Raspberry Pi Camera Integration Handover

10 inch FPV drone payload and analog video path

Field	Current record
Prepared	22 September 2026
Aircraft project	Rekon10 10 inch quadcopter on the existing 4S system
Payload	Raspberry Pi Zero 2 W and Raspberry Pi Camera Module 2
Purpose	Continue camera, analog-video, and power integration without losing verified settings or safety gates

Current conclusion. The Pi and camera work on the bench, and the Pi is configured to generate an NTSC composite preview automatically. The remaining gates are to finish and inspect the Pi-to-flight-controller wiring, choose a power source with adequate margin, and prove the complete image path in the goggles under a sustained propeller-off load test. The system is not yet cleared for flight.

Status at a glance

Item	Status	Evidence or remaining work
Pi operating system	Verified	Raspberry Pi OS Lite 64-bit based on Debian 13 Trixie
Camera hardware	Verified	IMX219 Camera Module 2 detected; 3280 x 2464 still capture passed
Remote access	Verified previously	SSH key login worked at the last known address; the Pi was offline when this handover was generated
Composite output	Configured	NTSC output enabled and a continuous 640 x 480, 30 fps DRM preview service installed
Pi video pads	User reported complete	Wires soldered to Pi TV and GND pads; solder quality and continuity not independently inspected
FC and VTX	Partially complete	VTX is reported soldered to the LUX F765; Pi wire terminations at FC Camera/GND are not verified
Pi power	Open decision	Matek MBEC6S candidate is only 1.5A continuous; FC 5V rail is shared and requires load-margin validation
Goggle image and latency	Not tested	No confirmed live image, OSD overlay, dropout test, or end-to-end latency measurement
Flight readiness	Not cleared	Power, RF, video, and hover revalidation remain incomplete

Verified Pi configuration

Setting	Value
Hostname	quadpi

Setting	Value
Username	ebach
Last known IPv4 address	192.168.40.193, assigned by DHCP
Wi-Fi MAC	88:A2:9E:C4:B8:13
Wi-Fi	RoboticsLan on 2.4 GHz; the Windows computer may use RoboticsLan-5G on the bridged subnet
Camera	Sony IMX219, Camera Module 2
Last successful camera test	3280 x 2464 JPEG; no corruption observed
Last successful power check	vcgencmd get_throttled returned 0x0
Camera software	rpicam-apps-lite; rpicam-hello available

Composite preview software

Boot configuration changes

- In /boot/firmware/config.txt: dtoverlay=vc4-kms-v3d,composite
- In /boot/firmware/cmdline.txt: vc4.tv_norm=NTSC
- Original files preserved as config.txt.pre-fpv and cmdline.txt.pre-fpv

Automatic preview service

```
quadpi-camera-preview.service
```

```
/usr/bin/rpicam-hello --timeout 0 --fullscreen --viewfinder-width 640 --viewfinder-height 480 --framerate 30
```

The service was enabled, reboot-tested, and observed active. Its journal reported a DRM preview window and the 640 x 480 IMX219 stream.

Required wiring

The Pi camera is converted to analog composite video by the Pi. The flight controller then inserts its analog OSD before sending video to the Rush Tank Solo VTX.

Connection	From	To	State
Camera data	Camera Module 2	Pi CSI connector through the 15-pin-to-22-pin ribbon	Connected and tested
Composite signal	Pi TV test pad	LUX F765 pad labelled Camera	Pi end soldered; FC end unverified
Video reference	Pi GND	LUX F765 GND	Pi end soldered; FC end unverified
Processed video	LUX F765 VTX pad	Rush Tank Solo video input	User reports VTX already soldered
VTX RF	Rush Tank Solo MMCX	Rated 5.8 GHz antenna	Must be attached before VTX power

Connection	From	To	State
Optional control	LUX TX4 S/A	Tank Solo SmartAudio	Not required to obtain an image

Video path. Camera Module 2 -> CSI ribbon -> Pi -> TV composite output -> FC Camera input -> FC OSD -> FC VTX output -> Tank Solo -> goggles

Power decision and risk

Preferred design. Use a dedicated 4S-input regulator providing a measured 5.0 to 5.1V at 3A continuous or greater. This isolates the flight-video computer from the flight controller's accessory rail and preserves current margin.

Rejected candidate. The photographed Matek MBEC6S has a selectable 5V or 9V output but is rated for only 1.5A continuous. It may boot the Pi under light load, but it does not meet the planned 3A continuous margin and should not be used for flight-critical video.

Proposed FC-powered alternative. The boxed LUX F765 pads labelled 5V and GND are electrically the correct pair for Pi power. However, the LUX manual rates the entire onboard 5V regulator at 2.5A, shared with the flight controller and other 5V equipment. This configuration couples a Pi overload or short to the flight-control power system and must not be accepted without a sustained load test.

Next safe milestone

Complete a propeller-off bench validation of the entire camera power and video chain. Do not proceed directly to a hover.

1 Inspect and meter with all power removed

1. Remove every propeller and disconnect the flight battery, USB cables, and CanaKit supply.
2. Photograph both sides of the Pi and FC solder joints under good light and magnification.
3. Confirm the Pi TV wire terminates only at the LUX Camera pad, not Camera Ctrl, VTX, a UART, or a power pad.
4. Confirm the Pi ground terminates at LUX ground and that no loose strand can contact an adjacent pad.
5. Check for an unintended short between Pi TV and GND and between Pi 5V and GND.
6. Verify that the VTX antenna is attached before any VTX power is applied.

2 Validate the selected power source

7. Power the regulator or FC without the Pi connected and measure polarity at the intended Pi leads.
8. Confirm 5.0 to 5.1V. Stop immediately for 9V, battery voltage, reversed polarity, unstable readings, or unexpected heating.
9. Connect the Pi only after the unloaded voltage and polarity pass.
10. Never connect the CanaKit adapter and airborne 5V source simultaneously.

3 Run the sustained propeller off test

11. Use a smoke stopper for the first low-load aircraft power-up. Keep the VTX at its minimum practical bench power.
12. Verify that the FC, receiver, Pi, camera service, VTX, and goggles all start without resets.
13. Measure voltage at the Pi while the camera preview is active. Treat voltage near or below 4.8V as inadequate margin.
14. Run the complete system for at least 15 minutes while watching the goggle image and checking component temperature.
15. Check vcgencmd get_throttled before and after the test. Any nonzero undervoltage history requires correction before flight.

16. Move wiring gently and exercise the radio controls while watching for image breakup, Pi reboot, receiver dropout, or FC reset.

4 Verify the pilot image

17. Confirm correct camera orientation, usable field of view, stable NTSC sync, and visible FC OSD.
18. Measure practical camera-to-goggle latency and compare it with the course's accepted setup before using it as the pilot view.
19. Confirm that the image recovers automatically after a normal power cycle.
20. Record the VTX channel, power, antenna, measured Pi voltage, test duration, temperatures, and any dropouts.

Stop conditions

Stop immediately if	Required response
BEC or FC output is not a stable 5V	Disconnect the Pi and correct the power source before continuing
Pi reports undervoltage or reboots	Increase power margin, shorten or enlarge power wiring, and repeat the full test
FC or receiver resets when the Pi starts	Do not use the shared FC rail; install a dedicated regulator
VTX has no antenna attached	Remove power and attach the correct antenna before retrying
Image rolls, flickers, disappears, or has severe motor-noise sensitivity	Inspect grounds and routing; do not fly from the feed
Any solder bridge, exposed conductor, smoke, odor, or rapid heating is observed	Remove power immediately and inspect under magnification

Remote access and service commands

The last known address is DHCP-assigned and may change. Check the router or scan for MAC 88:A2:9E:C4:B8:13 if 192.168.40.193 stops responding.

```
ssh -i C:\Users\ebach\.ssh\id_ed25519_rpi ebach@192.168.40.193
```

Purpose	Command on the Pi
Preview service status	systemctl status quadpi-camera-preview.service
Preview service log	journalctl -u quadpi-camera-preview.service -b --no-pager
Restart preview	sudo systemctl restart quadpi-camera-preview.service
List detected cameras	rpicam-hello --list-cameras
Check undervoltage history	vcgencmd get_throttled
Check temperature	vcgencmd measure_temp
Clean shutdown	sudo shutdown -h now

Known maintenance item

A full operating-system upgrade was attempted but stopped safely during package download because the Raspberry Pi repository slowed to a multi-hour estimate. No package installation had begun, and the package database was clean afterward. Complete updates later on reliable power and network service, then retest the camera service after reboot.

Records to preserve

- Clear photographs of Pi TV, GND, 5V, and power-ground joints and the matching FC terminations.
- Power-source make, model, current rating, output selection, measured unloaded voltage, and loaded voltage at the Pi.
- FC 5V rail loads if the shared rail is used.
- Goggle model, receiver module, VTX channel and power, antenna type, image orientation, and observed latency.
- Fifteen-minute test results, temperatures, get_throttled value, and any reset or dropout.

References

- Raspberry Pi Zero 2 W test pads: https://datasheets.raspberrypi.com/rpizero2/raspberry-pi-zero-2-w-test-pads.pdf
- Raspberry Pi composite output configuration: https://www.raspberrypi.com/documentation/computers/config_txt.html
- Raspberry Pi camera software: https://www.raspberrypi.com/documentation/computers/camera_software.html
- Lumenier LUX F765 NDAA manual: https://getfpv-media.s3.us-east-1.amazonaws.com/wysiwyg/files/Lumenier_LUX_F765_FC_NDAA_User_Manual_v1.pdf
- Matek MBEC6S product page: https://www.mateksys.com/?portfolio=mbec6s